

Problem 8. Consider the subspace

$$W = \text{span} \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^3$$

- (a) Find a basis (not necessarily orthonormal) for $W^\perp$.
- (b) Argue that together with the vector spanning W , these vectors form a basis $\mathfrak{B}$ for $\mathbb{R}^3$, and find the matrix B for orthogonal projection onto W with respect this basis.
- (c) Find the matrix A for orthogonal projection onto W with respect to the standard basis, and write down an equation of the form

$$B = S^{-1}AS$$

for an appropriately chosen S , showing that A and B are similar. No need to explicitly compute S^{-1} or the matrix products here, the equation should come from the fact that B and A represent the same transformation with respect to different bases.

Solution. (a) We can write $W = \text{im} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, so $W^\perp = \ker \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$. This matrix is already in reduced row echelon form. It corresponds to an equation $x - 2y + z = 0$. This yields $x = 2y - z$, and so a general element of the kernel has the form

$$\begin{bmatrix} 2y - z \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Our basis is

$$v_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

- (b) Let

$$w = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

This vector spans W . To show that w, v_1, v_2 are a basis for $\mathbb{R}^3$, let $u \in \mathbb{R}^3$ be any vector. Then u can be decomposed in a unique way as $u_1 + u_2$, where $u_1 \in W$, $u_2 \in W^\perp$. We can then write $u_1 = aw$ for a unique a , and $u_2 = bv_1 + cv_2$ for unique b and c . Thus shows that $u = aw + bv_1 + cv_2$ is a linear combination of w, v_1, v_2 , and moreover that the coefficients are uniquely determined by u . Because this holds for any u , w, v_1, v_2 are a basis for $\mathbb{R}^3$.

The projection onto W transforms these vectors as follows:

$$\text{proj}_W(w) = w$$

$$\text{proj}_W(v_1) = \text{proj}_W(v_2) = 0$$

Therefore, the matrix representation of proj_W in the basis w, v_1, v_2 is

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(c) We can find the matrix for projection in the standard basis by first normalizing w :

$$u = \frac{1}{|w|}w = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

then using the formula

$$A = uu^\top = \frac{1}{6} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1/6 & -1/3 & 1/6 \\ -1/3 & 2/3 & -1/3 \\ 1/6 & -1/3 & 1/6 \end{bmatrix}$$

To get an equation $B = S^{-1}AS$ we should take the columns of S to be our basis w, v_1, v_2 . Explicitly, the equation is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1/6 & -1/3 & 1/6 \\ -1/3 & 2/3 & -1/3 \\ 1/6 & -1/3 & 1/6 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$